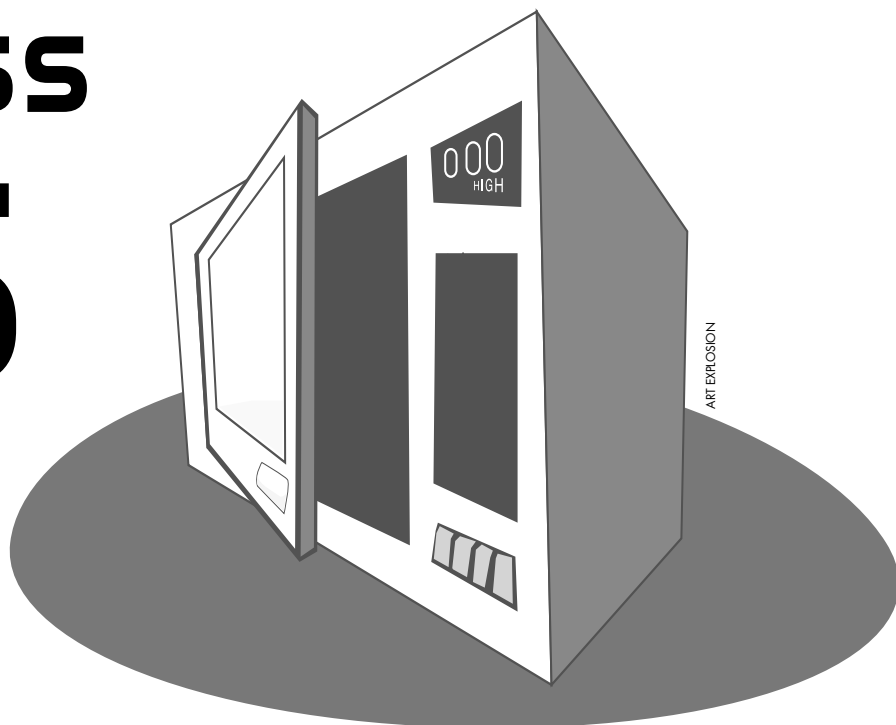


Cooking with Less GAS

by Jennifer Mitchell-Jackson and

Alan Meier



Microwave ovens, when they're on, stack up well against traditional ovens.

In today's fast-track society, more and more people are eating out, buying prepared foods, or converting conventional recipes for quick cooking in the microwave oven. Modern times require modern cooking devices like the microwave oven, and it turns out this modern approach to cooking is often the most energy-efficient way to make dinner.

Cooking four medium potatoes takes about 15 minutes in the microwave oven. In a conventional gas or electric oven, it would take about an hour. In addition to saving time, microwaves save energy because they are much more efficient than traditional ovens. Cooking potatoes in a microwave uses about 10%–25% of the energy used by a conventional oven, and results in much smaller heat losses (see Figure 1). Furthermore, a

microwave does not require preheating.

Not all foods are microwave friendly, but many foods are. Manufacturers of packaged foods often leave the cooking method up to the customer. When it comes to using a microwave versus an electric oven, this choice is easy. Cooking a frozen dinner in a microwave rather than in an oven saves almost 1 kWh of electricity. Choosing to microwave four boneless breasts of chicken, four potatoes, and stuffing for the family can save 2 kWh a day (if you can tolerate the taste and texture). Over a year, these savings can add up. If the average price of electricity is 10¢/kWh, saving 2 kWh per day will mean \$73 in energy savings per year.

A microwave wins on energy use but what about energy cost? When comparing a microwave to a gas oven, differing fuel costs can make

the decision a trickier one. Although a microwave is much more efficient and will cook foods faster, the price of gas is usually about one-fifth of the price of electricity. (These figures do not take into account current regional energy shortages.) One hour in a gas oven costs approximately the same as 30 minutes in a microwave. In the case of the four potatoes, therefore, it's still both more energy efficient and cheaper to choose the microwave.

Microwaves can also be used in place of an electric or gas cooktop. Instead of frying an egg or boiling water on a burner, many people turn to the microwave. An electric stove, however, is much more efficient than a microwave—70% compared to 55%—because the heat is applied directly to the container that is being used for cooking (see Figure 2). For example, when boiling water for a large mug of hot chocolate, an electric stove uses about 25% less electricity than a microwave oven. A gas stove is less efficient (40%) and requires more energy than a microwave; however, gas ranges usually cost less to use, due to the lower

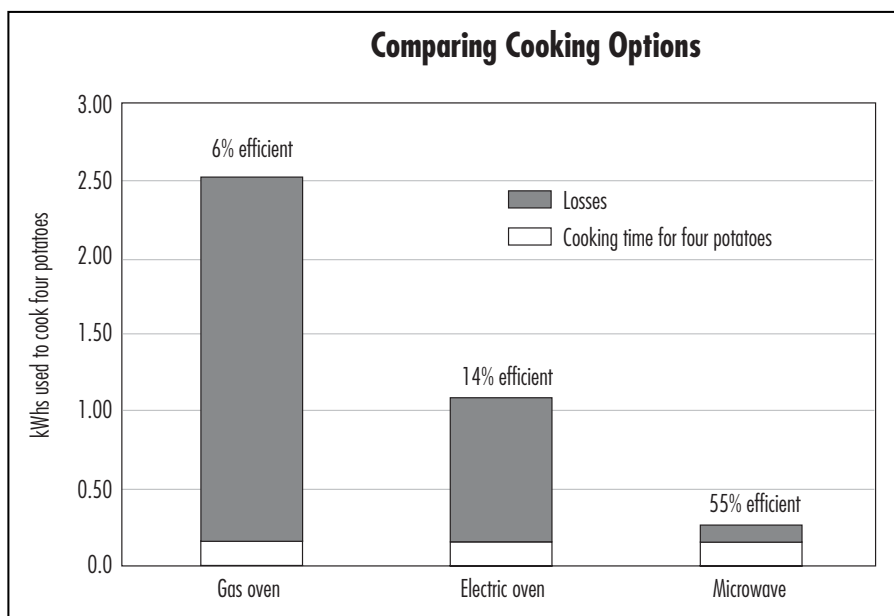


Figure 1. You can cook several potatoes in a microwave oven in about 15 minutes.

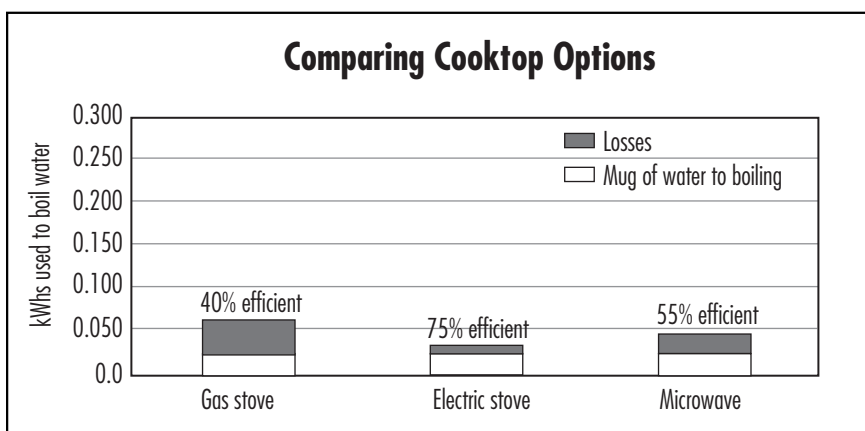


Figure 2. An electric stove is the best option for heating a cup of water for hot chocolate.

ally cost less to use, due to the lower price of natural gas (see Figure 2).

Even when the stove looks like a better option than the microwave, consumers can sometimes save more energy by using the microwave, since they tend to heat smaller portions in a microwave. Consider, for instance, boiling water for tea. A tea drinker would be likely to fill a single mug of water to heat in the microwave, but might fill a kettle to boil water on the stove (then go away and leave it until the water had boiled off).

Microwaves are also frequently used for defrosting. Here, thinking ahead is the key to saving money. Although defrosting is done at a lower power than cooking, putting that

frozen steak in the defrost compartment of the refrigerator won't cost a cent.

The Most Efficient Microwave Is Easy to Find

To find out just how efficient microwave ovens are, we tested ten different models using a procedure that closely resembled the International Electrotechnical Commission's standard method for measuring microwave oven performance. We borrowed the ovens from local households so that we could test a microwave population representative of those currently in use (see Table 1).

We found that the microwaves'

average efficiencies (the ratio of the electricity that went to heat food or water in the microwave divided by the total electricity used) ranged from 49% to 57%. These results were comparable to those from a 1994 study by Lawrence Berkeley National Laboratory (LBNL) which measured microwave efficiencies at 48%–60%. While we did not specifically test for the effects of size on efficiency, we noticed that the two largest microwaves were the least efficient. We also found that the newer ovens were generally more efficient than the older ovens, although our sample size was too small to determine if this finding was statistically significant.

Almost all of the power required by a microwave oven goes to the magnetron—a special electron tube that produces the microwaves. A combination of small components, such as a fan, a turntable, an electronic panel, a clock, and a light, draws a much smaller amount of power. Note, however, that small components such as the clock may end up consuming more energy over time than the magnetron, since the small components are on all the time (see “Your Microwave May Be Using More Energy Off Than On,” p. 36). Microwave efficiency, therefore, is mostly dependent on the efficiency of the magnetron, which is generally pretty high—usually above 70%.

Go Ahead, Have Your Popcorn

According to the Popcorn Institute, Americans consume over 17 billion quarts of popcorn each year. This means that the average American household consumes approximately three bags of popcorn a week. If half of this popcorn is popped in a microwave oven, which cooks in about three minutes per bag, then the average household uses only about 4 kWh per year to cook popcorn. At a price of 10¢/kWh, that's only 40¢ per household per year in energy costs. Preparing popcorn on a stovetop is particularly inefficient because of low heat transfer.

Your Microwave May Be Using More Energy Off Than On

Estimates indicate that the 84 million microwave ovens in U.S. homes on average are used about 1½ hours per week, or 78 hours per year. For the remaining 8,682 hours in the year, however, many of these units still consume energy.

In a study conducted by LBNL last year, microwaves with electronic panels or clocks were found to use an average of 2.8 watts even when they were not in use. Since microwaves are usually plugged in all the time, a microwave that draws 2.8 watts in standby mode would use approximately 24 kWh per year.

Field measurements of 18 microwave ovens in New Zealand recorded significant energy use when the microwaves were in standby mode. Standby power accounted for about



Electronic panels or clocks that are on all the time can waste electricity.

40% of all the electricity used, and six of the microwaves consumed more electricity in a year while in standby mode than while cooking.

While the 1 to 3 watts continually used by one microwave oven may seem small, a typical U.S. house has dozens of appliances. Added together, these appliances can use up to 50 watts of standby power. Furthermore, standby power consumption may grow as microwaves are upgraded to include new sensors and features. It is important for manufacturers, home energy efficiency professionals, and homeowners to make sure that these new features are not drawing additional power when in standby mode since this energy consumption adds up to a significant draw on national energy production.

offers the promise of improvements in the future.

See the Savings

Between reheating food, microwaving dinners, boiling water, and defrosting, the typical residential microwave oven consumes about 110 kWh of electricity per year, and costs about \$11.00 a year to operate. Relying more on the microwave may increase this

cost, but it can cut down on cooking energy costs overall.

To streamline your kitchen's energy consumption, bear in mind these five tips:

- Limit the amount of food being cooked to the portion that will be consumed.
- Always consider using a microwave oven rather than an electric or gas oven.
- Think ahead and defrost in the

refrigerator rather than in the microwave.

- Don't worry about which brand to purchase, since the range of microwave efficiencies tends to be small.
- Consider pulling the plug when your microwave is not in use for long periods.



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Table 1. A Comparison of Microwave Efficiencies

Model	Size	Year	Average % Efficiency
Samsung MW5480W	small	1999	57.4
General Electric JE740WY 001	small	1998	56.2
Goldstar MA 7802	small	1997	56.4
Samsung MW4350W	small	1996	55.5
Sharp Carousel II	large	1993	49.5
Goldstar MA 4002	tiny	1991	54.3
Emerson AR 734	small	1986	54.8
Amana ML451T	small	1986	50.8
Panasonic	large	1986	48.9
Tappan 500 M56-1036	small	1985	52.3

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